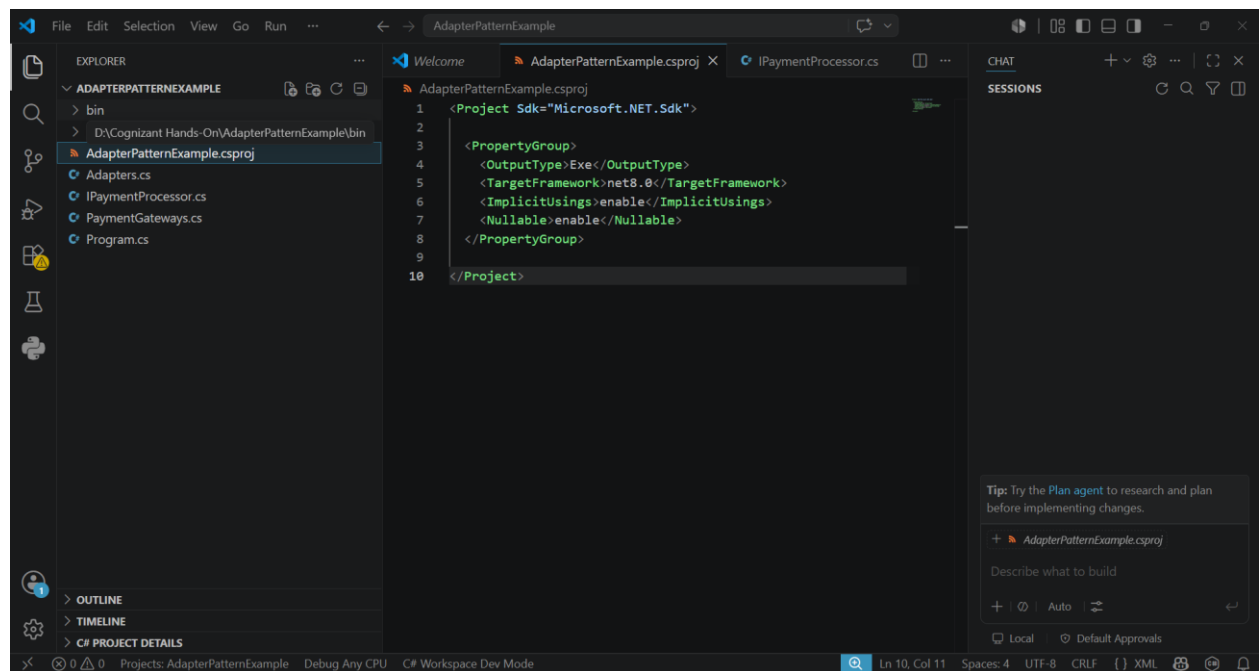


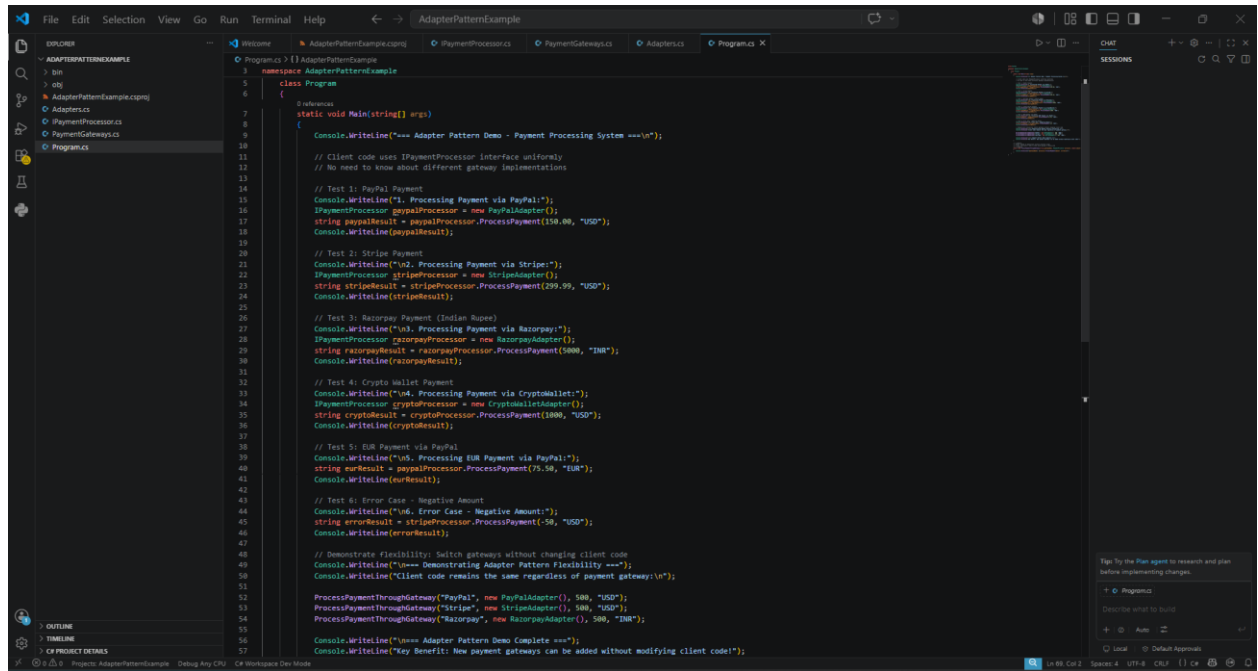
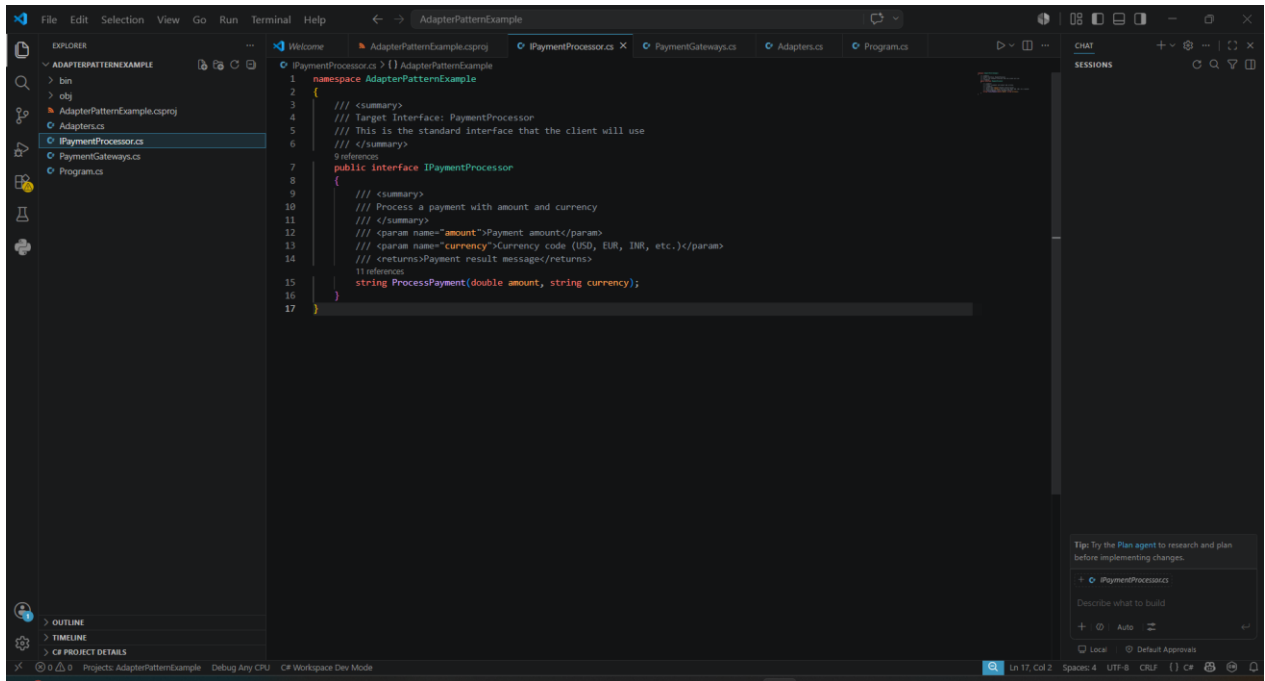
Hands-On

Exercise 4: Implementing the Adapter Pattern

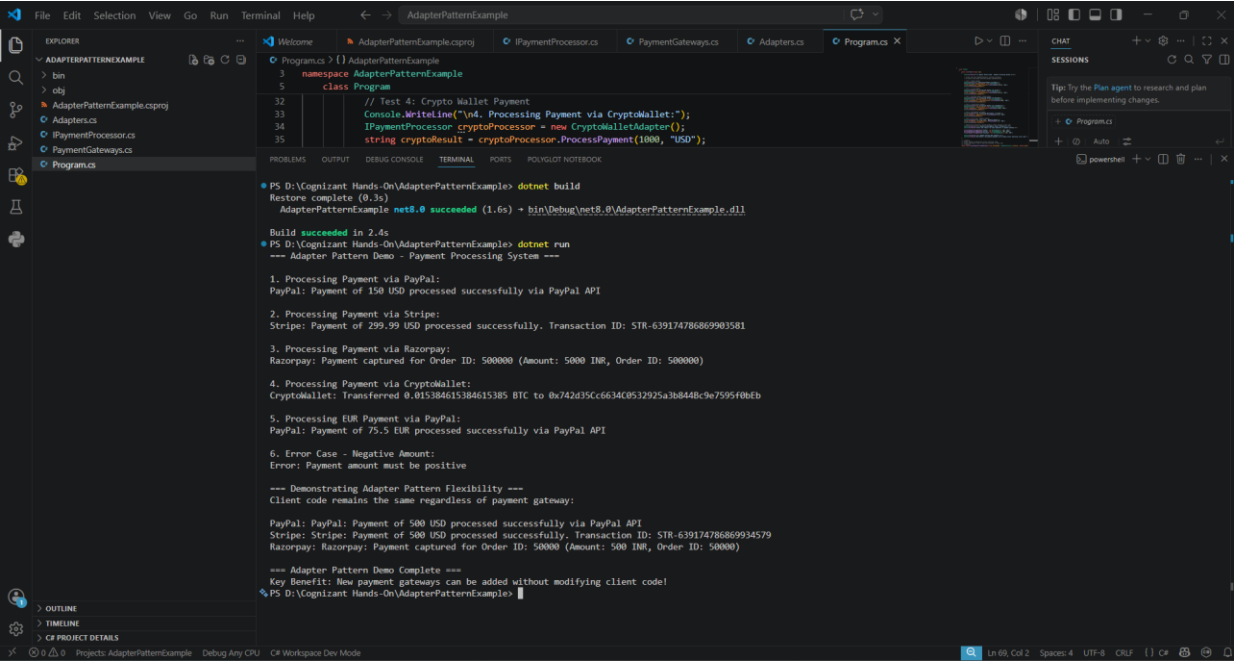
In this module, I successfully implemented, executed, and tested the Adapter Pattern in .NET by creating a project called AdapterPatternExample. I developed a PaymentProcessor interface that acts as the common target interface and created different payment gateway classes, each having its own methods and interfaces. I then created adapter classes that implement the PaymentProcessor interface and convert payment requests into calls to the respective gateway methods, allowing seamless integration of these gateways. After implementation, I executed the program and tested various payment gateways using their corresponding adapters to ensure that all of them could be used interchangeably through a common interface. This practical exercise helped me understand the importance of the Adapter Pattern in making incompatible interfaces work together, improving code reusability, and simplifying the integration of third-party services into .NET applications.

Coding:





Output:



```
File Edit Selection View Go Run Terminal Help
AdapterPatternExample

Programs > {} AdapterPatternExample
3 namespace AdapterPatternExample
5 class Program
32 // Test 4: Crypto Wallet Payment
33 Console.WriteLine("v4. Processing Payment via CryptoWallet:");
34 IPaymentProcessor cryptoProcessor = new CryptoWalletAdapter();
35 string cryptoResult = cryptoProcessor.ProcessPayment(1000, "USD");

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POLYGLOT NOTEBOOK

PS D:\Cognizant_Hands-On\AdapterPatternExample> dotnet build
Restore complete (0.1s)
AdapterPatternExample net8.0 succeeded (1.6s) → bin\Debug\net8.0\AdapterPatternExample.dll

Build succeeded in 2.4s
PS D:\Cognizant_Hands-On\AdapterPatternExample> dotnet run
--- Adapter Pattern Demo - Payment Processing System ---

1. Processing Payment via PayPal:
PayPal: Payment of 150 USD processed successfully via PayPal API

2. Processing Payment via Stripe:
Stripe: Payment of 299.99 USD processed successfully. Transaction ID: STR-639174786869903581

3. Processing Payment via Razorpay:
Razorpay: Payment captured for Order ID: 500000 (Amount: 5000 INR, Order ID: 500000)

4. Processing Payment via CryptoWallet:
CryptoWallet: Transferred 0.015384615384615385 BTC to 0x742d35Cc6634C0532925a3b844bC7e7595f0b4b

5. Processing EUR Payment via PayPal:
PayPal: Payment of 75.5 EUR processed successfully via PayPal API

6. Error Case - Negative Amount:
Error: Payment amount must be positive

--- Demonstrating Adapter Pattern Flexibility ---
Client code remains the same regardless of payment gateway:

PayPal: PayPal: Payment of 500 USD processed successfully via PayPal API
Stripe: Stripe: Payment of 500 USD processed successfully. Transaction ID: STR-639174786869934579
Razorpay: Razorpay: Payment captured for Order ID: 500000 (Amount: 500 INR, Order ID: 500000)

--- Adapter Pattern Demo Complete ---
Key Benefit: New payment gateways can be added without modifying client code!
PS D:\Cognizant_Hands-On\AdapterPatternExample>
```